Construcciones tipo Lebeau-Robbiano

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Recap: Setting

$$\dot{\phi}(t) = A\phi(t), \quad \phi(0) = x \in \mathcal{E}, \quad t \geq 0, \tag{1}$$

$A : D(A) \subset \mathcal{E} \to \mathcal{E}$ is the generator of a strongly continuous semigroup $(e^{tA})_{t\geq 0}$ on a Hilbert space $\mathcal{E}$.
observation operator $C \in \mathcal{L}(D(A), \mathcal{F})$ admissible for this semigroup, i.e, C is a continuous operator from $D(A)$ with the graph norm to another Hilbert space $\mathcal{F}$ and satisfies

$$\int_0^T \|Ce^{tA}x\|^2 dt \leq \mathrm{Adm}_T \|x\|^2, \qquad x \in D(A), \quad T > 0. \tag{2}$$

We say that (A, C, T) is observable at cost $\kappa_T > 0$ if

$$\|e^{TA}x\|^2 \leq \kappa_T \int_0^T \|Ce^{tA}x\|^2 dt, \qquad x \in D(A). \tag{3}$$

Recap: Principal lemma

We are interested in the asymptotic growth of κ_T as $T \to 0$ and think of κ_T as the cost of fast control.

Lemma (Lemma 2.1)

If the approximate observability estimate

$$f(t)\|e^{tA}x\|^2 - f(qt)\|x\|^2 \leq \int_0^t \|Ce^{\tau A}x\|^2 d\tau, \qquad x \in D(A), \quad t \in (0, T'], \tag{4}$$

holds with $f(t) \to 0$ as $t \to 0^+$, $q \in (0,1)$ and $T' > 0$, then $\kappa_T \leq 1 f((1-q)T)$ for $T \in (0, T']$, i.e, the fast control cost does not grow more than the inverse of f.

Recap: Relative observability on growth subspaces

$e^{tA}\mathcal{E}_\lambda \subset \mathcal{E}_\lambda \subset \mathcal{E}_{\lambda'}$ for $t > 0$ and $\lambda' > \lambda$. This family satisfies the growth condition

$$\|e^{tA}x\| \leq m_0 e^{-\lambda t}\|x\|, \quad x \perp \mathcal{E}_\lambda, \quad t \in (0, T_0), \quad \lambda > 0. \tag{5}$$

We also think them as spectral subspaces of A

$$\sigma(A\,|_{\mathcal{E}_\lambda^\perp}) \subset \{z \in \sigma(A) \mid \Re z \leq -\lambda\}.$$

We also assume that there is an admissible observation operator C_0 satisfying the bound relative to C on growth spaces with exponent $\alpha \in (0, 1)$ and rate $a > 0$,

$$\|C_0 x\|^2 \leq a_0 e^{2a\lambda^\alpha}\|Cx\|^2, \qquad x \in \mathcal{E}_\lambda, \quad \lambda > 0. \tag{6}$$

When the reference operator C_0 satisfies the observability cost estimate with exponent $\beta > 0$ and rate $b > 0$,

$$\|e^{TA}x\|^2 \leq b_0 e^{\frac{2b}{T^\beta}} \int_0^T \|C_0 e^{tA}x\|^2 dt, \quad x \in D(A), \quad T \in (0, T_0), \tag{7}$$

Recap: principal result

Theorem (Theorem 1)

Under the assumptions (5), (6) *and* (7) *with* $\beta = \frac{\alpha}{1-\alpha}$, *the system* (A, C, T) *is observable at a cost* κ_T *such that* $2c = \limsup_{T\to 0} T^{\beta} \ln \kappa_T < \infty$. *More precisely, this rate* c *is bounded in terms of an implicitly defined* $s > 0$*:*

$$c \leq c_* := \left(\frac{b}{a}(\beta+1)\right)^{\frac{\beta+1}{\beta}} \frac{\beta^{\beta}}{s^{\frac{(\beta+1)^2}{\beta}}}, \quad s(s+\beta+1)^{\beta} = (\beta+1)\beta^{\frac{\beta^2}{\beta+1}} \frac{b^{\frac{1}{\beta+1}}}{a}. \tag{8}$$

Moreover, if the admissibility constant in (2) *satisfies* $\mathrm{Adm}_T \to 0$ *as* $T \to 0$, *then there exists* $T' > 0$ *such that* $\kappa_T \leq 4a_0 b_0 \exp\left(\frac{2c_*}{T^{\beta}}\right)$ *for* $T \in (0, T']$.

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Admissibility

Any $C \in (\mathcal{E}, \mathcal{F})$ satisfies the admissibility condition (2) with $\mathrm{Adm}_T = T\|C\|^2$. It might be useful to circumvent the admissibility assumption

Lemma

The conclusion $2c = \limsup_{T\to 0} T^\beta \ln \kappa_T \leq 2c_*$ *of Theorem 2 is still valid if we replace the assumption that* $C \in \mathcal{L}(D(A), \mathcal{F})$ *satisfies the admissiblity condition* (2) *by the following time smoothing effect assumption*

$$\forall x \in \mathcal{E}\ \forall t > 0,\ e^{tA}x \in D(A), \quad \limsup_{t\to 0} t^\beta \ln \|Ae^{tA}\| = 0. \tag{9}$$

Controllability cost

From the definition of κ_T in (3), we have $\kappa_T \leq \|e^{(T-T')A}\|^2 \kappa_{T'}$ for $T' < T$. This justifies why $\sqrt{\kappa_T}$ does not grow more than the semigroup as $T \to +\infty$.

Moreover, if $\kappa_t \leq f(t)$ for $t \in (0, T']$ with g non-increasing, then $\kappa_t \leq Mg(t)$ for $t \in (0, T]$ with $M = \sup_{s\in(0,T-T')} \|e^{sA}\|^2 < \infty$. This justifies that we restrict to some bounded intervals $(0, T']$ in the statement of our results.

The dual problem to the final-observability is the null-controllability of

$$\dot{f}(t) = A^* f(t) + Bu(t), \qquad f(0) = f_0 \in \mathcal{E}, \ t \geq 0, \tag{10}$$

with input $u \in L^2([0, T]; \mathcal{F})$ and control operator $B = C^* \in \mathcal{L}(\mathcal{F}, D(A^*))$. Since C satisfies the admissibility condition (2), B satisfies

$$\left\| \int_0^T e^{tA^*} Bu(t) \right\|^2 \leq K_T \int_0^T \|u(t)\|^2 dt,$$

Controllability cost pt. 2

and the solution of (10) is

$$f(T) = e^{TA^*} f_0 + \int_0^T e^{(T-t)A^*} Bu(t)dt.$$

More precisely, if (A, C, T) is observable at cost κ_T, then, for all f_0, there is a u such that $f(T) = 0$ and $\int_0^T \|u(t)\|^2 dt \leq \kappa_T \|f_0\|^2$.

Integrated observability estimate

Lemma 2.1 can be seen as the discrete version of the following lemma which has been used with $f(t) = \exp(-c/t)$ when proving observability by some parabolic global Carleman estimate

Lemma

If the integrated observability estimate

$$\int_0^T f(t)\|e^{tA}x\|^2 dt \leq \int_0^T \|Ce^{tA}x\|^2 dt, \quad x \in D(A), \quad T \in (0, T_0), \quad (11)$$

holds with $T_0 > 0$ and f is an increasing function such that $f(t) \to 0$ when $t \to 0^+$, then $\kappa_T \leq M_T/f(T)$ for $T \in (0, T_0)$ with $M_T = \sup_{t\in[0,T]} \|e^{tA}\|^2 \leq M_{T_0} < \infty$, i.e, the fast control cost does not grow more than the inverse of f. Conversely, if final-observability (3) holds for $T \in (0, T_0)$, then integrated observability (11) holds for $T \in (0, T_0)$ with $f(T) = 1/(T_0\kappa_T)$.

Approximate observability

The following lemma clarifies the connection of Lemma 2.1 and the approximate controllability.

Lemma

If (A, C, T) *satisfy the approximate observability estimate,*

$$\|e^{TA}x\|^2 \leq \kappa \int_0^T \|Ce^{tA}x\|^2 dt + \varepsilon \|x\|^2, \quad x \in D(A), \tag{12}$$

for some positive κ *and* ε*, then approximate null-controllability of* (10) *holds, i.e, for all* f_0 *there exists* u *such that* $\|f(T)\|^2 \leq \varepsilon \|f_0\|^2$ *and* $\int_0^T \|u(t)\|^2 dt \leq \kappa \|f_0\|^2$.

Normal semigroups and spectral spaces

Normal semigroups have a spectral decomposition in terms of a projection-valued measure E. We have

$$E_\lambda = E(\{z \in \sigma(A) \mid \Re z > -\lambda\}), \qquad \mathcal{E}_\lambda = E_\lambda \mathcal{E}.$$

If there's an orthonormal basis $\{e_n\}$ of $\mathcal{E}$, then $\mathcal{E}_\lambda = \mathrm{Span}\{e_n\}_{\lambda_n < \lambda}$. (5) becomes an estimate on sums of eigenfunctions of A.

When $A = -\Delta$ on $\mathcal{E} = L^2(\mathbb{R}^d)$, the spectral decomposition is the Fourier transform

$$\widehat{f(-\Delta)\phi}(\xi) = f(|\xi|)^2 \hat{\phi}(\xi), \quad \phi \in L^2(\mathbb{R}^d),$$

thus $\phi \in \mathcal{E}_\lambda$ just means $\hat{\phi}(\xi) = 0$ for $|\xi|^2 > \lambda$, i.e, ϕ is the restriction to eh real axis of an entire function $\tilde{\phi}$ such that $|\tilde{\phi}(z)| \leq c e^{\sqrt{\lambda}|\Im z|}$ by the Paley-Wiener theorem.

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Anomalous diffusions

M smooth, connected Riemannian manifolds with boundary. Δ denotes the Laplace-Beltrami with domain $H_0^1(M) \cap H^2(M)$.
$\mathcal{E} = \mathcal{F} = L^2(M)$, $\mathcal{E}_\lambda$ are the spectral spaces, $C_0 = \mathrm{Id}$ and $C = \mathbf{1}_\Omega$, a non-empty subset.
For $A = -(-\Delta)^\gamma$ and $\gamma > 1/2$, the reference operator estimate (6) holds with $\alpha = \frac{1}{2\gamma}$, so the Theorem 2.2 says

Theorem

For all $\gamma > 1/2$, the anomalous diffusion:

$$\partial_t \phi + (-\Delta)^\gamma \phi + \mathbf{1}_\Omega u, \quad \phi(0) = \phi_0 \in L^2(M),\ u \in L^2([0,T] \times M),$$

is null-controllable in any time $T > 0$, Moreover, the cost κ_T satisfies $\limsup_{T \to 0} T^\beta \ln \kappa_T < \infty$ *with* $\beta = \frac{1}{2\gamma - 1}$.

Structural damping

A self-adjoint with bounded inverse in H Hilbert. Let H_λ be the spectral spaces of A corresponding to $\{z \in \sigma(A) \mid \Re z < \lambda\}$. Consider observation operators $C_0, C : \mathcal{L}(D(A), \mathcal{F})$ satisfying the main assumption

$$\|C_0 z\|^2 \leq d_0 e^{2d\lambda^\delta} \|Cz\|^2, \quad z \in H_\lambda, \quad \lambda > 0. \tag{13}$$

Here we consider $B = C^*$ and

$$\ddot{\zeta}(t) + \rho A^{2\gamma} \dot{\zeta}(t) + A^2 \zeta(t) = Bu(t), \ (\zeta(0), \dot{\zeta}(0)) = (\zeta_0, \zeta_1) \in D(A) \times H,$$

and $u \in L^2([0, T], \mathcal{F})$ with structural dissipation $\gamma \in (0, 1)$. We can write as

$$A = \begin{pmatrix} 0 & I \\ -A^2 & -\rho A^{2\gamma} \end{pmatrix}, \quad D(A) = \{(z_0, z_1) \in \mathcal{E} \mid Az_0 + \rho A^{2\gamma - 1} z_1 \in D(A)\}.$$

Structural damping pt. 2

By Lumer-Phillips, $-A$ generates a contraction semigroup.

Theorem

Recall that δ and d are the exponent and rate in the main assumption (13). For all $\rho > 0$ and $\gamma \in (\delta/2, 1-\delta/2)$, for all ζ_0, ζ_1, there is an input u such that the solution ζ satisfies $\zeta(T) = \dot{\zeta}(T) = 0$ and the cost estimate

$$\int_0^T \|u(t)\|^2 dt \leq b_0 \exp\left(\frac{2b}{T^\beta}\right)(\|A\zeta_0\|^2 + \|\zeta_1\|^2),\ \zeta_0 \in D(A),\ \zeta_1 \in H,$$

where T is small and

$$\beta = \left(\frac{2}{\delta}\min\{\gamma, 1-\gamma\} - 1\right)^{-1}, \qquad b > \frac{d^{\beta+1}}{\min\{\rho/2, 1/\rho\}^\beta} \frac{(\beta+1)^{\beta(\beta+1)}}{\beta^{\beta^2}}.$$

Structural damping pt. 3

Main application: Plate equation with square root damping and interior control in Ω with hinged boundary condition on M:

$$\begin{cases} \ddot{\zeta} - \rho\Delta\dot{\zeta} + \Delta^2\zeta = \mathbf{1}_\Omega u, & [0,T]\times M, \\ \zeta = \Delta\zeta = 0, & [0,T]\times \partial M, \\ \zeta(0) = \zeta_0 \in H^2(M)\cap H_0^1(M), & \\ \dot{\zeta}(0) = \zeta_1 \in L^2(M), & \\ u \in L^2([0,T]\times M). & \end{cases} \tag{14}$$

Applying the previous theorem to $A = -\Delta$ with $\delta = \gamma = 1/2$, we get a better result than Miller 2006 Thm 1, improving the value of β.



L. Miller (2006)
The cost of fast non-structural controls for a linear elastic system with structural damping

.

Structural damping pt. 4

Under the geometrical optics condition from Bardos-Lebeau-Rauch, that the length L_Ω of the longest generalized geodesic in $\overline{M}$ which does not intersect Ω is not ∞, the second part of Miller 2006 Thm 2 estimates the cost rate: for all $\rho \in (0, 2)$, the control cost of the plate eq. satisfies the previous estimate with $\beta = 1$ and any $b > b_1 L_\Omega^2$ for some b_0, b_1 which do not depend on Ω and ρ.



L. Miller (2006)
The cost of fast non-structural controls for a linear elastic system with structural damping

.

Diffusion in a potential well

Consider the Schrödinger operator $A = \Delta - V = \Delta + |x|^{2k}$ for $x \in \mathbb{R}^d$ and $k \in \mathbb{N}$, which is negative, self-adjoint and has compact resolvent on

$$D(A) = \{\phi \in H^2(\mathbb{R}^d) \mid \|V\phi\|_{L^2} < \infty\}.$$

Consider $\Gamma = \{x : |x| > r_0,\ x/|x| \in \Omega_0\}$ for $\Omega_0 \subset S^{d-1}$. Also use $\mathcal{E} = \mathcal{F} = L^2(\mathbb{R}^d)$, spectral spaces as growth spaces, $C_0 = \mathrm{Id}$ and $C = \mathbf{1}_\Gamma$. Then the reference operator estimate (6) is proved in Miller 2008 with $\alpha = \frac{1}{2}(1 + \frac{1}{k})$, with some eigenfunctions concentrating at an equator satisfying

$$\|e_n\|^2 \geq a_0 e^{2a\lambda_n^\alpha} \|Ce_n\|^2.$$



L. Miller (2008)
Unique continuation estimates for sums of semiclassical eigenfunctions and null-controllability from cones

.

Diffusion in a potential well pt. 2

Theorem

For all $k > 1$, the diffusion in the potential well $V(x) = |x|^{2k}$:

$$\partial_t \phi - \Delta \phi + V\phi = \mathbf{1}_\Gamma u, \ \phi(0) = \phi_0 \in L^2(\mathbb{R}^d), \ u \in L^2([0,T] \times \mathbb{R}^d),$$

is null-controllable in any time $T > 0$. Moreover, the cost κ_T satisfies $\kappa = \limsup_{T \to 0} T^\beta \ln \kappa_T < \infty$ with $\beta = 1 + \frac{2}{k-1}$.
If there is a vector space of dimension 2 in $\mathbb{R}^d$ which does not intersect the closure $\overline{\Omega_0}$, then $\kappa \neq 0$.

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Rate for the eigenfunction estimate

$A = \Delta$ Dirichlet Laplacian, (M, g) compact, smooth, connected. $\mathcal{E} = \mathcal{F} = L^2(M)$ and $\mathcal{E}_\lambda$ spectral space. $C_0 = \mathrm{Id}$ and $C = \mathbf{1}_\Omega$, a subset of $\overline{M}$.

Theorem

If the observability estimate

$$\int_M |\phi(x)|^2 dx \leq c_0 e^{c\sqrt{\lambda}} \int_\Omega |\phi(x)|^2 dx,$$

holds for any $\lambda > 0$ and any sum of eigenfunctions

$$\phi = \sum_{\mu \in \sigma(-\Delta) \cap (0,\lambda)} \varphi_\mu, \qquad -\Delta \varphi_\mu = \mu \varphi_\mu,$$

then $c \geq \sup_{y \in M} \mathrm{dist}(y, \overline{\Omega})$.

Proof pt. 1

The proof relies on an estimate of the heat kernel in small time. By finite propagation for wave eq. and Kannai's formula

$$e^{t\Delta} = \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^{+\infty} e^{-\frac{-s^2}{4t}} \cos(s\sqrt{-\Delta}) ds,$$

we get using C (resp. C') as the multiplication by $\mathbf{1}_\Omega$ (resp. $\mathbf{1}_{\Omega'}$), that

$$\|Ce^{t\Delta}C'\|_{\mathcal{L}(L^2)} \leq \frac{1}{\sqrt{4\pi t}} \int_{|s|\geq\rho} e^{-\frac{-s^2}{4t}} ds \leq e^{-\frac{\rho^2}{4t}}. \tag{15}$$

Where $\rho = \operatorname{dist}(\Omega, \Omega')$. Now we use functional calculus, let $\mathbf{1}_{\{-\Delta<\lambda^2\}}$ be a projection-valued measure. Fix a normalized eigenfunction φ_μ. UCP for elliptic operators implies that $Y = \{y \in M \setminus \overline{\Omega} : \varphi_\mu(y) \neq 0\}$ is an open dense in $M \setminus \overline{\Omega}$.

Proof pt. 2

Now we can take supremum in Y instead of M. Eigenfunctions are continuos, so we can find a small ball Ω' with center y s.t

$$\left|\int_{\Omega'} \varphi_\mu(x)dx\right| > 0, \qquad \left|\text{dist}(\Omega,\Omega') - \text{dist}(y,\overline{\Omega})\right| < \varepsilon.$$

Now by the observability estimate,

$$\|\phi\| \leq a_0 e^{a\lambda}\|C\phi\|, \quad \phi \in \mathbf{1}_{\{-\Delta<\lambda^2\}}(L^2(M)),$$

we want to prove $2a \geq \rho = \text{dist}(\Omega,\Omega')$, with $\chi = \mathbf{1}_{\Omega'}$ s.t $|\langle\chi,\varphi_\mu\rangle| > 0$. Consider $\phi_\lambda = e^{\frac{\rho}{2\lambda}\Delta}\chi$ and $\phi = \mathbf{1}_{-\Delta<\lambda^2}\phi_\lambda$. By definition of C', $C'\chi = \chi^2 = \chi$, hence

$$\|C\phi_\lambda\| \leq e^{-\frac{\rho\lambda}{2}}\|\chi\|.$$

Since $\phi_\lambda - \phi = \mathbf{1}_{-\Delta\geq\lambda^2}e^{\frac{\rho}{2\lambda}\Delta}\chi$, the spectral theorem yields

$$\|C(\phi_\lambda - \phi)\| \leq \|\phi_\lambda - \phi\| \leq e^{-\frac{\rho\lambda}{2}}\|\chi\|.$$

Proof pt. 3

Adding these inequalities yields $\|C\phi\| \leq 2e^{-\frac{\rho\lambda}{2}}\|\chi\|$. Since $\|\varphi_\mu\| = 1$ an its an eigenfunction, we also have

$$\|\phi\| \geq |\langle \phi, \varphi_\mu \rangle| = |\langle \chi, e^{\frac{\rho}{2\lambda}\Delta}\varphi_\mu \rangle| = e^{-\frac{\rho\mu^2}{2\lambda}}|\langle \chi, \varphi_\mu \rangle|.$$

Plugging this inequalities in the observability estimate and using that $|\langle \chi, \varphi_\mu \rangle| > 0$,

$$0 < e^{-\frac{\rho\mu}{2}}|\langle \chi, \varphi_\mu \rangle| \leq 2a_0 e^{(a-\rho/2)\lambda}\|\chi\|, \quad \lambda \geq \mu.$$

Taking the limit as $\lambda \to \infty$ shows that $a < \rho/2$ is forbidden. □